4.3.2 Properties of the sine function

From the graph of $y = \sin x$, we observe the following properties of sine function:

(i) There is no break or discontinuities in the curve. The sine function is continuous.

(ii) The sine function is odd, since the graph is symmetric with respect to the origin.

(iii) The maximum value of sine function is 1 and occurs at $x = \cdots, -\frac{3\pi}{2}, \frac{\pi}{2}, \frac{5\pi}{2}, \cdots$ and the minimum value is -1 and occurs at $x = \cdots, -\frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}, \cdots$. In otherwords, $-1 \le \sin x \le 1$ for all $x \in \mathbb{R}$.

4.3.3 The inverse sine function and its properties

The sine function is not one-to-one in the entire domain $\mathbb{R}$. This is visualized from the fact that every horizontal line $y = b,\ -1 \le b \le 1,$ intersects the graph of $y = \sin x$ infinitely many times. In other words , the sine function does not pass the horizontal line test, which is a tool to decide the **one-to-one** status of a function. If the domain is restricted to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then the sine function becomes one to one and onto **(bijection)** with the range $[-1, 1]$. Now, let us define the inverse sine function with $[-1, 1]$ as its domain and with $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ as its range.

Definition 4.3

For $-1 \le x \le 1,$ define $\sin^{-1} x$ as the unique number y in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $\sin y = x$. In other words, the inverse sine function $\sin^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is defined by $\sin^{-1}(x) = y$ if and only if $\sin y = x$ and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Note

(i) The sine function is one-to-one on the restricted domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, but not on any larger interval containing the origin.

(ii) The cosine function is non-negative on the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, the range of $\sin^{-1} x$. This observation is very important for some of the trigonometric substitutions in Integral Calculus.

(iii) Whenever we talk about the inverse sine function, we have,

$\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$ and $\sin^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

(iv) We can also restrict the domain of the sine function to any one of the intervals, $\cdots \left[-\frac{5\pi}{2}, -\frac{3\pi}{2}\right], \left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right], \left[\frac{\pi}{2}, \frac{3\pi}{2}\right], \left[\frac{3\pi}{2}, \frac{5\pi}{2}\right] \cdots$ where it is one-to-one and its range is $[-1, 1]$.

(vi) The restricted domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is called the **principal domain** of sine function and the values of $y = \sin^{-1} x, -1 \le x \le 1,$ are known as **principal values** of the function $y = \sin^{-1} x$.

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